cables *continued*
underground 23
underwater 187
caissons 95
cameras
brain implants 243
digital 152–153, 158
driverless cars 47
keyhole surgery 238–239
optical mouse 166, 167
optical sorting 222
robots 174, 175
smartphones 206
speed 50–51
television 188
cams 122, 123
camshafts 42
cancer 84, 234, 235
cantilever bridges 94
cantilever cranes 103
capacitive touch screens 204, 206
capillary action 218
carbon-fiber polymers 80–81, 241
carbon nanotubes 84
cardboard 82–83, 223
cars
biofuel 31
driverless 47, 171, 175
electric and hybrid 10, **46–47**
gas-engine supercars 10
hydrogen-powered 34, 35
internal combustion engines 42–43
working of **44–45**
cast iron 73
caster effect 41
casting 73, 74
catalysts 78, 79
catamarans 55
caterpillar tracks 92
catheters 232, 246, 247
cathode-ray tube (CRT) 190
cathodes 32, 33, 34, 35, 111, 190, 234
cell phones 186–187, 196, 202, 204, 206–207, 232
cell towers 185
celluloid 79
cement 76
center of gravity 41, 55
central heating 106, 108–109
central processing unit (CPU) 162, 164, 165
centrifugal force 118, 119
CGI imagery 177
chain reactions 24
Channel Tunnel 96
charged-coupled device (CCD) 133, 155
chemical energy 11, 42
chemical reactions 32, 34
chemical rockets 69
chess, computer 171
chips 161, 163
chlorination 13
chlorofluorocarbons (CFCs) 117
chromatograms 245
chromosomes 244
circuit breakers, magnetic 107
circuits, electrical 16, 17, 19, 125
cisterns 120, 121
cliff sensors 119
clothing **128–129**, 208
cloud computing, in agriculture 221
clutch 44–45
coordinates 158, 159
coagulation 12
coal-fired power stations 20, 21
cochlear implants 243
code
computer programs 169
locks and alarms 122, 123, 125
coffee machines 113
coffers 88
Cold War 192
collective controls 65
collimators 234
color filters 150, 191
combination boilers 108
combination locks 123
combine harvesters **216–217**
combustion chambers 60, 108
command & control (C&C) servers 200, 201
communications
electronic paper 208–209
email 200–201
communications *continued*
Internet 196–197
mobile devices 204–205
radio waves 180–181
radios 182–183
satellite navigation 194–195
satellites 186, 187, 192–193
smartphones 206–207
telecommunications networks 186–187
telephones 184–185
televisions 188–191
Wi-Fi 202–203
World Wide Web 198–199
commutators 17, 18, 19
compact discs 141
compact fluorescent lamps 144, 145
compilers 169
composites 76, **80–81**
composting toilets 121
compression 88, 89, 94
compressors 60, 116, 117
computer-generated imagery (CGI) 177
computers **162–165**
artificial intelligence 170–171
design 86
digital electronics 160–161
electrowetting displays 209
email 200–201
farming 220, 221
Internet 196–197
keyboards and mice 166–167
medical 233
printers and scanners 154–155
robots 172–175
smartphones 206–207
software 168–169, 196, 206
virtual reality 176–177
Wi-Fi 202–203
World Wide Web 198–199
concrete **76–77**, 81
condensers 21, 25, 116, 117
conduction 72
conductive layers 190
connectivity, in mobile devices 205
conservation of energy 11
construction
arches and domes 88–89
bridges 94–95
cranes 102–103
earth movers 92–93
elevators 100–101
skyscrapers 98–99
tunnels 96–97
contact sensors 125
control rods 24
cooking
kettles and toasters 112–113
microwave ovens 110–111
*see also* food
cooling towers 21
corrosion 74, 75
cotton 126, 127
counterjibs 102
counterweights 100, 101, 102
cows 214–215, 229
cracking 15
cranes **102–103**
cranks 40
crankshafts 42–43, 44
crashing, computers 162
credit cards 148
crops **212–213**, 218–219, 220–221, 222–223, 228
crude oil **14–15**, 78
CT scans 235
curtain walls 98
cyclic controls 65
cyclonic vacuum cleaners 119
cylinders
hydraulics 93
internal combustion engines 42–43
locks 122–123
cylindrical tumbler locks 122–123

# D

dairy farming 214–215
dams 28, 29
data 159, 165
data capture 205
data centers 197, 198
data exchange 196–197, 202–203, 205